

# **NETWORKING AND ROUTING PROTOCOL**

**ET 8315**

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**Lecture 2**

**ROUTER FUNDAMENTALS**

**College of Information and Communication Technology (CoICT)**

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# TOPICS

- Operating Cisco IOS Software
- Starting a Router

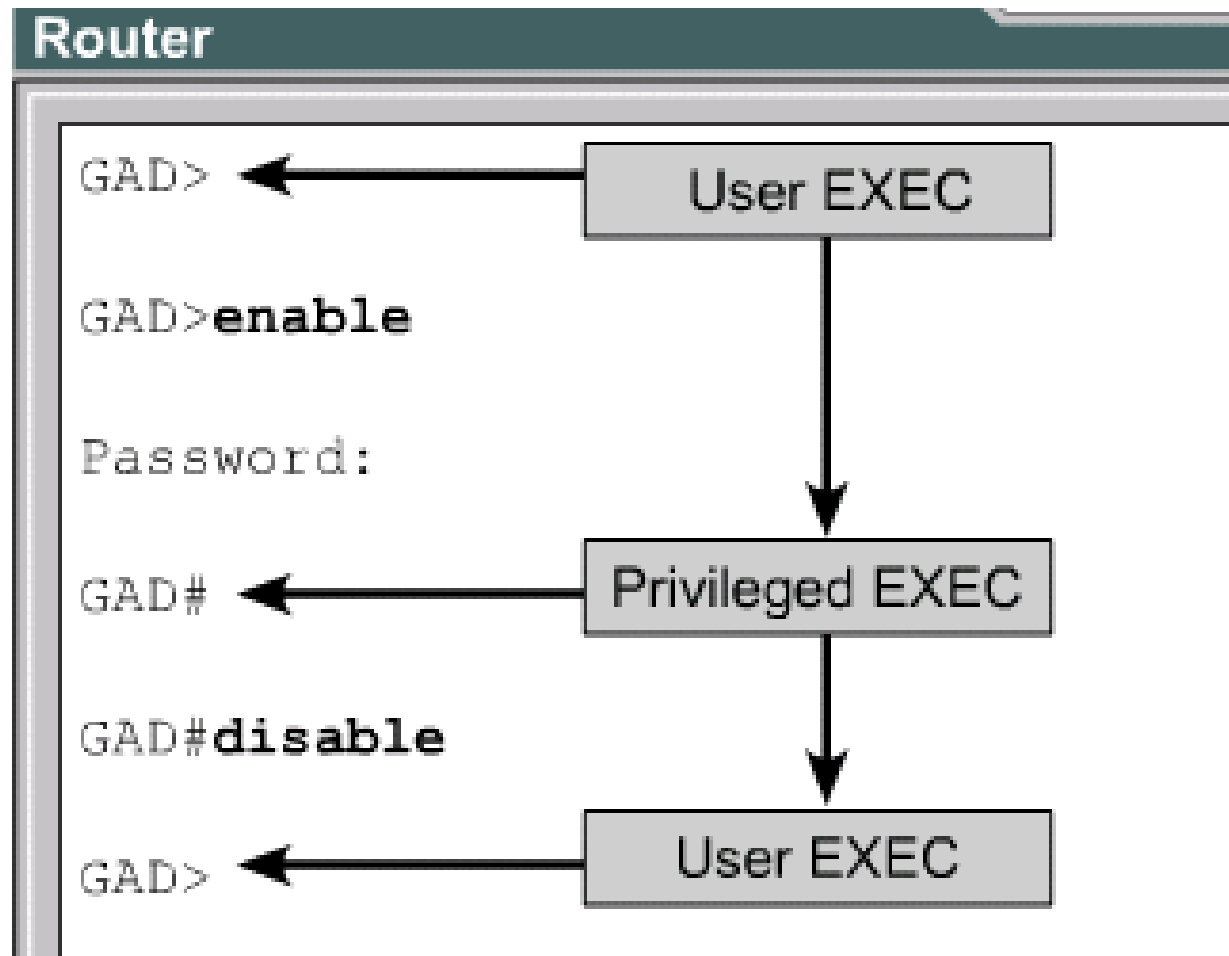
# The purpose of Cisco IOS software

- A **router** or **switch** cannot function without an OS.
- The Cisco Internetworking Operating System (IOS) provides the following network services:
  - ✓ Basic routing and switching functions
  - ✓ Reliable and secure access to networked resources
  - ✓ Network scalability
- The IOS software uses a command-line interface (CLI) as the traditional console environment. Access the CLI through:
  - ✓ A console session
  - ✓ A dialup connection using a modem
  - ✓ Telnet session to the router

# Router user interface modes

- The IOS provides a command interpreter service known as the **command executive (EXEC)**. The EXEC **validates** and **executes** the command
- The EXEC session is separated in **two 2 levels** of access
  - ✓ **User Exec**; view only mode to **check router status**
    - router>
  - ✓ **Privilege Exec**; also known as **enable** mode
    - router#
    - Enter the **enable** command at the “>” prompt
    - Enter configuration and management commands

# Router user interface modes



# Cisco IOS software features

- The **naming** convention for the different Cisco IOS releases contains three parts:
  - ✓ The **platform** on which the image runs
  - ✓ The **special features** supported in the image
  - ✓ **Where the image runs** and whether it has been zipped or compressed
- In general, the newer the release and the more features that it provides, the more memory it requires.
- Check **RAM** and **FLASH** for available memory.

# Checking Memory Requirements

```
BHM#show version
```

```
Cisco Internetwork Operating System Software  
IOS (tm) 1700 Software (C1700-BNSY-L), Version  
12.2(11)P, RELEASE SOFTWARE (fcl)
```

```
... <output omitted>...
```

```
System image file is "flash:c1700-y7-mz", booted via  
flash
```

```
cisco 1721 (68380) processor (revision C) with  
3584K/512K bytes of memory.
```

To find out the amount of flash memory:

```
GAD#show flash
```

```
... <output omitted>...
```

```
15998976 bytes total (10889728 bytes free)
```

# Operation of Cisco IOS Software

- The startup process of the router normally loads into RAM and executes one of 3 operating environments:
  - ✓ **ROM monitor**- performs the bootstrap process and provides low-level functionality and diagnostics. Used to recover from system failures and recover from a lost password. Available only through console.
  - ✓ **Boot ROM**- limited subset of the Cisco IOS. Allows write operations to flash memory and is used primarily to replace the Cisco IOS image that is stored in flash ex: copy tftp flash

# Operation of Cisco IOS Software

- ✓ Cisco IOS- stored in Flash, but loaded and executed from RAM

| Operating Environment | Prompt          | Usage                        |
|-----------------------|-----------------|------------------------------|
| ROM monitor           | > or ROMMON>    | Failure or password recovery |
| Boot ROM              | Router (boot) > | Flash image upgrade          |
| Cisco IOS             | Router>         | Normal operation             |

# Initial startup of Cisco routers

- The startup routines done to start the router operations must accomplish the following:
  - ✓ Make sure that the router hardware is tested and functional i.e. the CPU, memory, and interfaces
  - ✓ Find and load the Cisco IOS software.
  - ✓ Find and apply the startup configuration file or enter the setup mode.

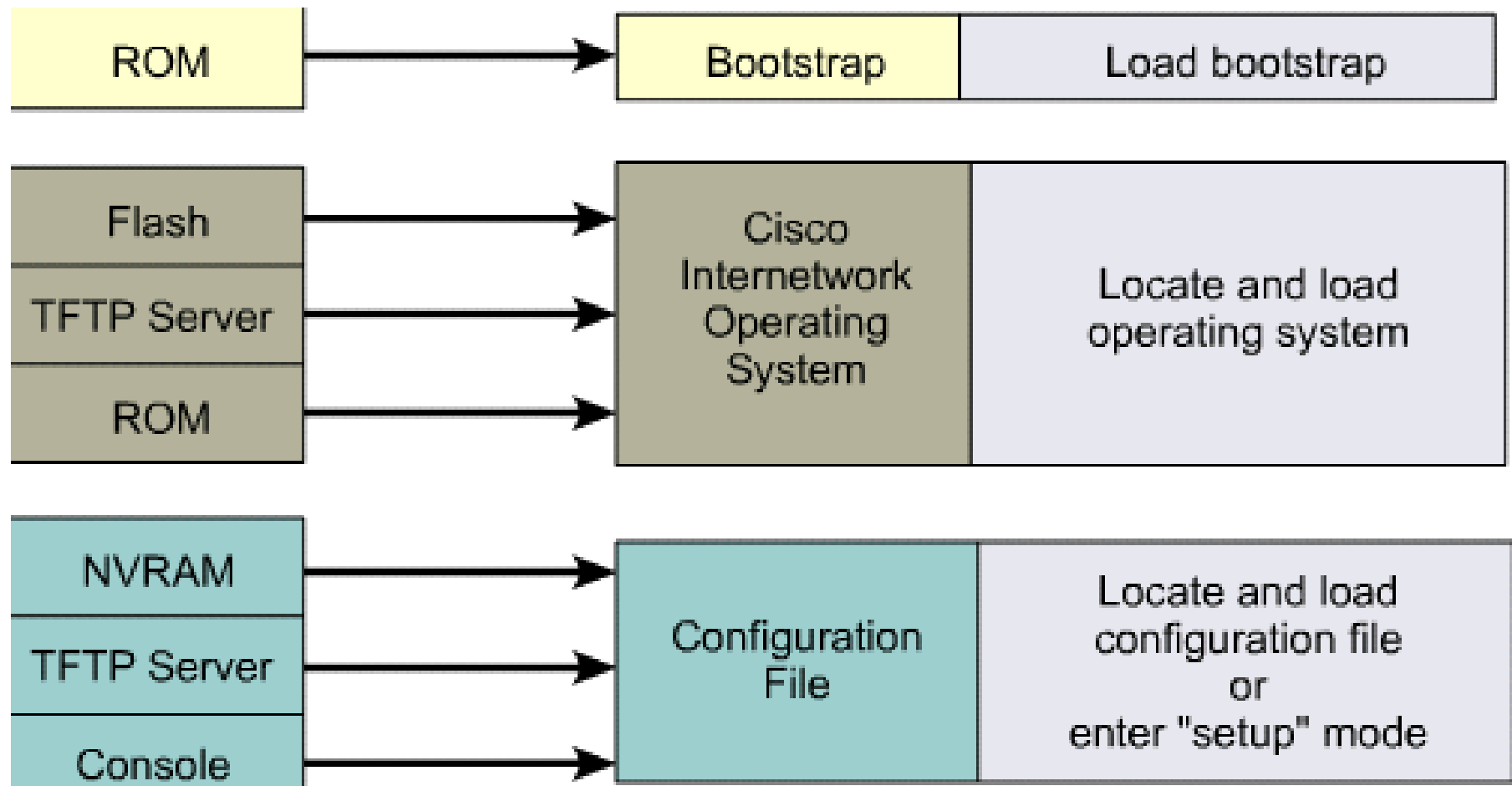
# Initial startup of Cisco routers

- After the POST, the following occur as the router initializes:
  - ✓ The generic bootstrap loader in ROM executes
    - The bootstrap loads instructions that cause other instructions to be loaded
  - ✓ The operating system is loaded
    - The location is disclosed in the boot field of the configuration register
  - ✓ The operating system locates the hardware and software components and lists the results on the console terminal

# Initial startup of Cisco routers

- ✓ The configuration file saved in NVRAM is loaded into main memory and executed one line at a time
  - The commands start routing processes, supply addresses for interfaces, and define other operating characteristics of the router
- ✓ If no configuration file is found, the operating system enters setup mode

# Initial startup of Cisco routers



# Initial startup of Cisco routers

- Setup is not intended as the mode for entering complex protocol features in the router.
- When a router cannot find its configuration file from any other source

# SETUP MODE

- Default answers appear in square brackets [ ] following the question. Press the **Enter** key to use these defaults.
- During the setup process, **Ctrl-C** can be pressed at any time to terminate the process.
- When setup is terminated using **Ctrl-C**, all interfaces will be administratively shutdown.

# SETUP MODE

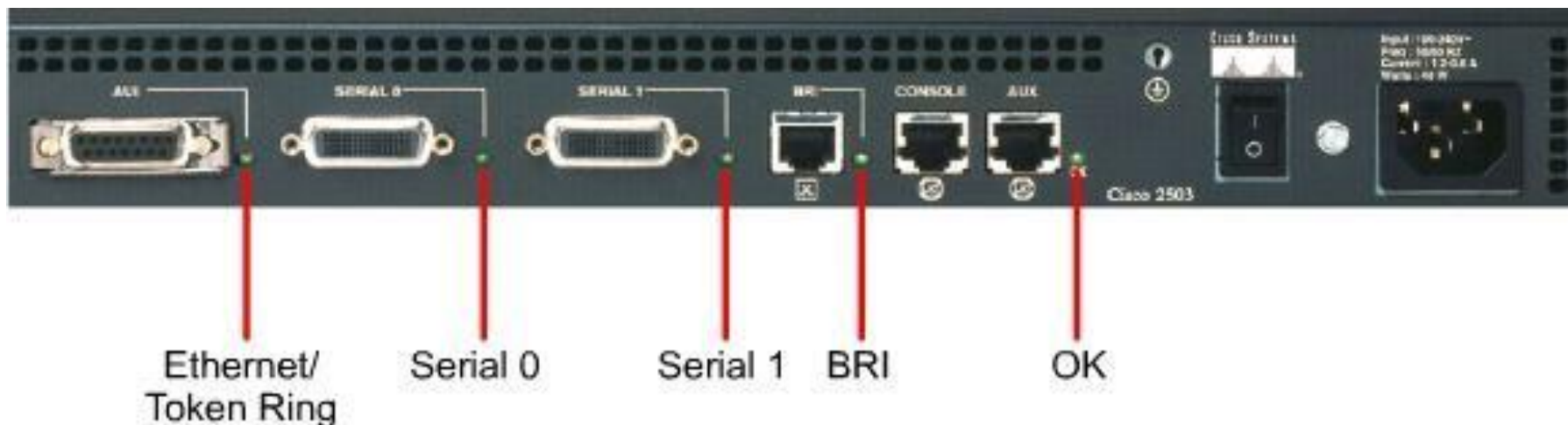
- When you complete the configuration process you will be prompted:
  - 1) Go to the IOS command prompt without saving this config
  - 2) Return back to the setup without saving this config
  - 3) Save this configuration to nvram and exit

# Startup and Boot sequence summary

- Checks hardware -POST
- Loads bootstrap
- Loads operating system
- Loads configuration file
  - ✓ Stored in NVRAM after setup
  - ✓ Also available from TFTP server
  - ✓ If not configured enters setup mode through the console

# Router LED indicators

- If a LED is off when the interface is active and is correctly connected, a problem may be indicated
- If an interface is extremely busy, its LED will always be on
- The green OK LED to the right of the AUX port will be on after the system initializes correctly



# Examining the initial router bootup

- **NVRAM invalid, possibly due to write erase**, tells the user that this router has not been configured yet or that the NVRAM has been erased.
- The user has the option to enter setup mode

```
Router
System Bootstrap, Version X.X(XXXX) [XXXXXX XX], RELEASE
SOFTWARE
Copyright (c) 1986-199X by Cisco Systems
2500 processor with 4096 Kbytes of main memory

Notice: NVRAM invalid, possibly due to write erase.

F3: 5797928+162396+258800 at 0x3000060

Restricted Rights Legend
```

```
Notice: NVRAM invalid, possibly due to write erase.
--- System Configuration Dialog ---

At any point you may enter a question mark '?' for help.
Refer to the 'Getting Started' Guide for additional help.
Use ctrl-c to abort configuration dialog at any prompt.
Default settings are in square brackets '[]'.
Would you like to enter the initial configuration dialog?
[yes]:
```

5/12/2014

# Examining the initial router bootup

- The output displayed during initial bootup includes:
- ✓ Bootstrap version
  - ✓ The IOS version
  - ✓ The router model,
  - ✓ The processor
  - ✓ The amount of memory the router contains NVRAM, RAM, and FLASH
  - ✓ The number of interfaces
  - ✓ The types of interfaces

```
Cisco Systems, Inc.  
170 West Tasman Drive  
San Jose, California 95134-1706  
Cisco Internetwork Operating System Software  
IOS (tm) X000 Software (XXX-X-X), Version XX.X(XXXX)  
[XXXXX XXX]  
Copyright (c) 1986-199X by Cisco Systems, Inc.  
Compiled Fri 20-Oct-9X 16:02 by XXXXX  
Image text-base: 0x03030FC0, data-base: 0x00001000  
Cisco 25XX (68030) processor (revision A) with 4092K/2048K  
bytes of memory.  
Processor board ID 00000000 X.25 software, Version X.X,  
NET2, BFE and GOSIP compliant.  
TN3270 Emulation software (copyright 1994 by TGV Inc).  
Basic Rate ISDN software, Version X.X.  
X Ethernet/IEEE 802.3 interface.  
2 Serial network interfaces.  
1 ISDN Basic Rate interface.  
32K bytes of non-volatile configuration memory.  
8192K bytes of processor board System flash (Read ONLY)
```

Same as **Show  
Version**  
command

# Establishing a HyperTerminal session

- Connect the terminal using:
  - ✓ an RJ-45 to RJ-45 rollover cable
  - ✓ an RJ-45 to DB-9 or RJ-45 to DB-25 adapter
- The parameters for the console port are:
  - ✓ 9600 baud,
  - ✓ 8 data bits,
  - ✓ no parity,
  - ✓ 1 stop bit,
  - ✓ No flow control. The console port does not support hardware flow control

# LOGGING INTO THE ROUTER

- When accessing a router, a user **must login** to the router before any other commands are entered.
- Router must be accessed with a terminal or by remote access
- **User EXEC mode** – allows the user to check the router status. **No router configuration changes are allowed**. Only a subset of the commands available at privileged EXEC level
- **Privileged EXEC mode** – allows the user to change the router configuration. Global configuration mode can be accessed along with other specific modes :
  - ✓ Interface or Subinterface
  - ✓ Line
  - ✓ Router
  - ✓ Route-map

# LOGGING INTO THE ROUTER

Router con0 is now available.

Press RETURN to get started.

User Access Verification

Password:

Router> ← User-Mode Prompt

Router>**enable**

Password:

Router# ← Privileged-Mode Prompt

Router#**disable**

Router>

Router>**exit**

Can type **ena**  
or **en** instead  
of "enable"

Two commands can be used to set a password used to access privileged EXEC mode:  
**enable password** and  
**enable secret**. If both commands are used, the **enable secret** command takes precedence

# Keyboard help in the router C L I

- Typing a **?** will display all command available at either modes.
- Typing in part of a command and a **?** will give all the options for completing that command
  - ✓ Ex: **cl?** Compare with **clock ?**
- Screen will only display **22** lines, and a **–MORE–** indicates that multiple screens are available as output
  - ✓ Enter key to view one line at a time
  - ✓ Space bar to view the next screen

# Keyboard help in the router CLI



Caret  
indicates  
place of error

```
Cisco>?
Exec commands:
access-enable      Create a temporary Access-List
                   entry
access-profile      Apply user-profile to interface
access-template    Create a temporary Access-List
                   entry
archive            manage archive files
bfe                For manual emergency modes
                   setting
cd                 Change current directory
clear              Reset functions
clock              Manage the system clock
configure           Enter configuration mode
connect            Open a terminal connection
copy               Copy from one file to another
--More--
```

```
Cisco#cl?
clear clock
Cisco#clock
% Incomplete command.
Cisco#clock ?
      set  Set the time and date
Cisco#clock set
% Incomplete command.
Cisco#clock set ?
      hh:mm:ss  Current Time
```

---

```
Cisco#clock set 19:50:00
% Incomplete command.
Cisco#clock set 19:50:00 ?
      <1-31>  Day of the month
      MONTH  Month of the year
Cisco#clock set 19:50:00 14 7
                                     ^
% Invalid input detected at '^' marker.
Cisco#clock set 19:50:00 14 July
% Incomplete command.
Cisco#clock set 19:50:00 14 July ?
      <1993-2035>  Year
Cisco#clock set 19:50:00 14 July 2003
Cisco#
```

# ENHANCED EDITING COMMANDS

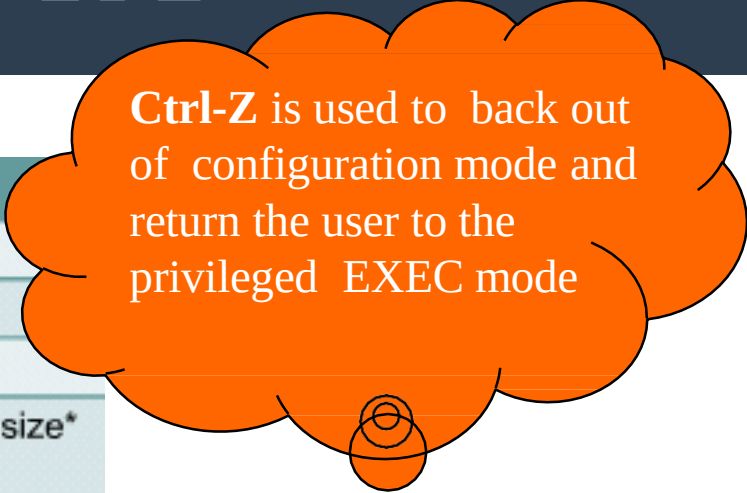
- Use “**Ctrl-A**” to move to the beginning of the line
- Use “**Ctrl-B**” to move the cursor back one character
- Use “**Ctrl-E**” to move to the end of the command line
- Enhanced editing is automatically enabled with the current software release
- You can disable enhanced editing (or the shortcut keys) by typing **terminal no editing**.
- When the cursor reaches the right margin, the command line shifts ten spaces to the left. A **\$** indicates that the line has been scrolled to the left.

# COMMAND HISTORY

- By default, the command history is enabled and the system records **10 command** lines.
- The maximum number of commands is 256.
- Use the **“terminal history size”** or the **“history size”** command to change the number of commands stored in the buffer
- Use **“Ctrl-P”** or the **up arrow** key repeatedly to recall successively older commands.
- **“Ctrl-N”** or the **down arrow** key recalls successively more recent commands.
- As a shortcut, you may enter the unique characters for a command, press the **“Tab”** key, and the interface will finish the entry for you.

# COMMAND HISTORY

| Command                                                  | Description                           |
|----------------------------------------------------------|---------------------------------------|
| <b>Ctrl-P</b> or up arrow key                            | Recalls last (previous) command       |
| <b>Ctrl-N</b> or down arrow key                          | Recalls most recent command           |
| Router> <b>show history</b>                              | Shows command buffer                  |
| Router> <b>terminal history size<br/>number-of-lines</b> | Sets the command history buffer size* |
| Router> <b>terminal no editing</b>                       | Disables advanced editing features    |
| Router> <b>terminal editing</b>                          | Re-enables advanced editing           |
| <b>&lt;Tab&gt;</b>                                       | Completes the entry                   |



**Ctrl-Z** is used to back out of configuration mode and return the user to the privileged EXEC mode

| Command                 | Description                                |
|-------------------------|--------------------------------------------|
| Ctrl-A                  | Moves to the beginning of the command line |
| Esc-B                   | Moves back one word                        |
| Ctrl-B (or right arrow) | Moves back one character                   |
| Ctrl-E                  | Moves to the end of the command line       |
| Ctrl-F(or left arrow)   | Moves forward one character                |
| Esc-F                   | Moves forward one word                     |

# Commands to check IOS

- The **show version** command:
  - ✓ IOS version and descriptive information
  - ✓ Bootstrap ROM version
  - ✓ Boot ROM version
  - ✓ Router up time
  - ✓ Last restart method
  - ✓ System image file and location
  - ✓ Router platform
  - ✓ Configuration register setting





**NEXT LECTURE**

**CONFIGURING A ROUTER**